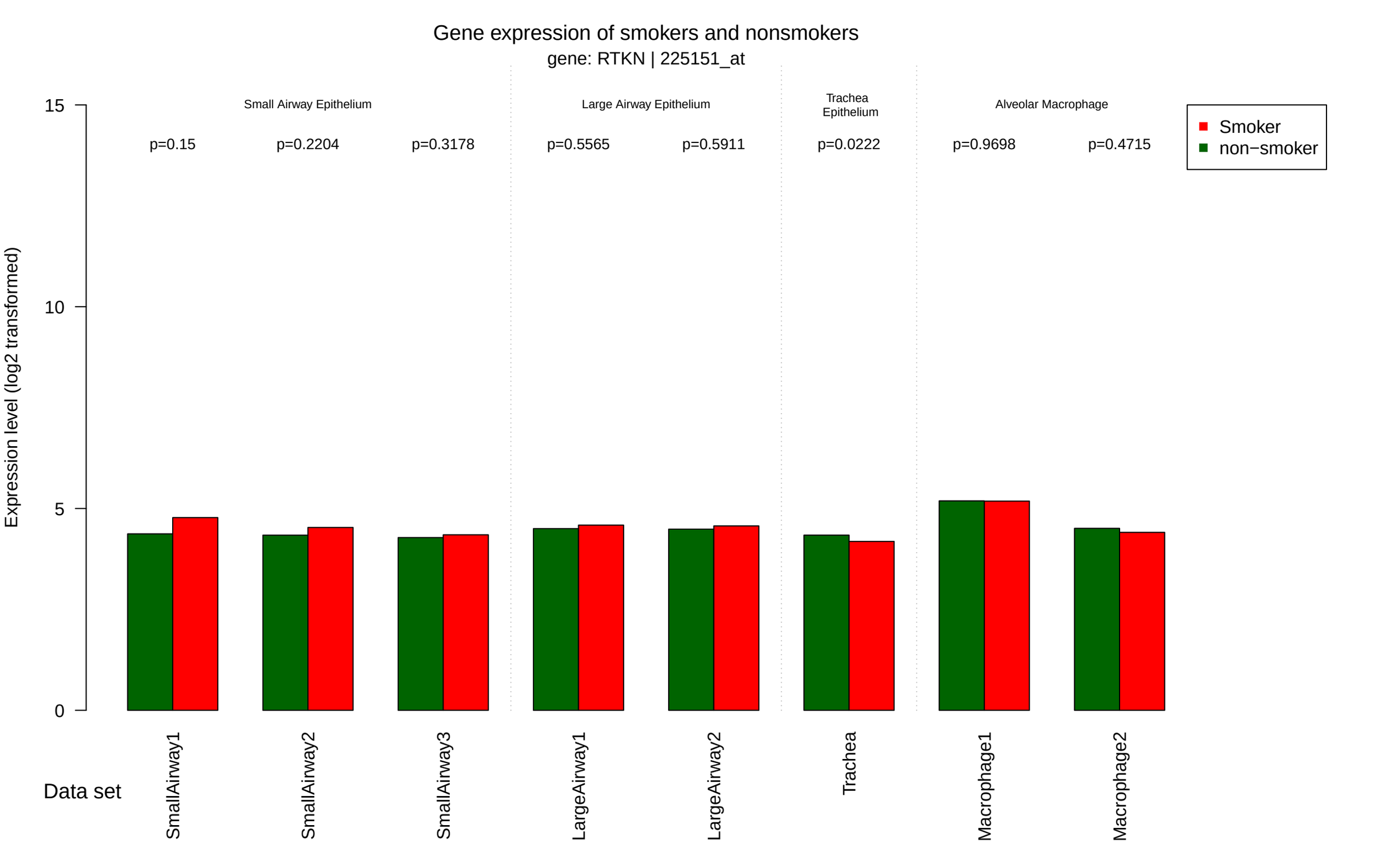
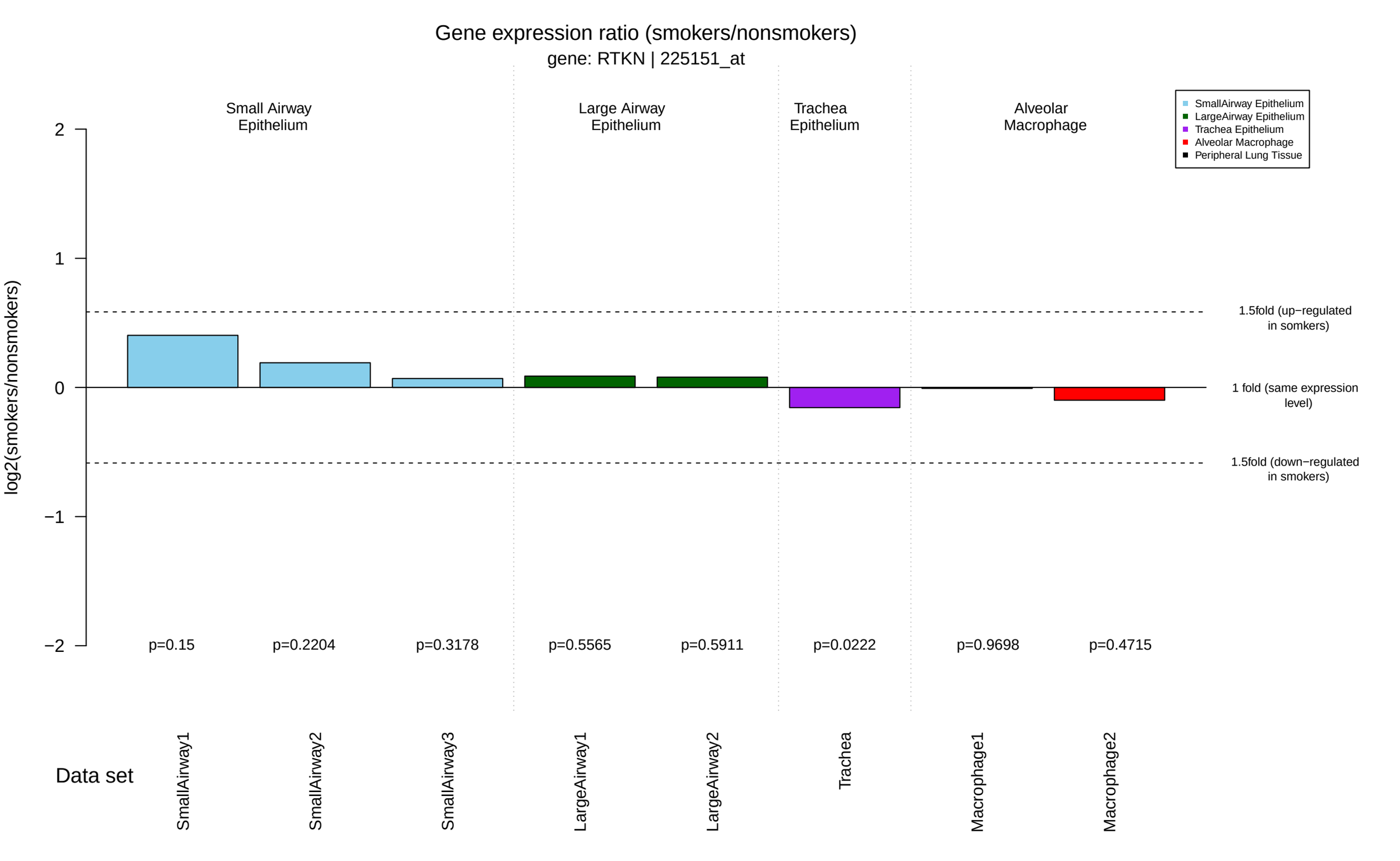
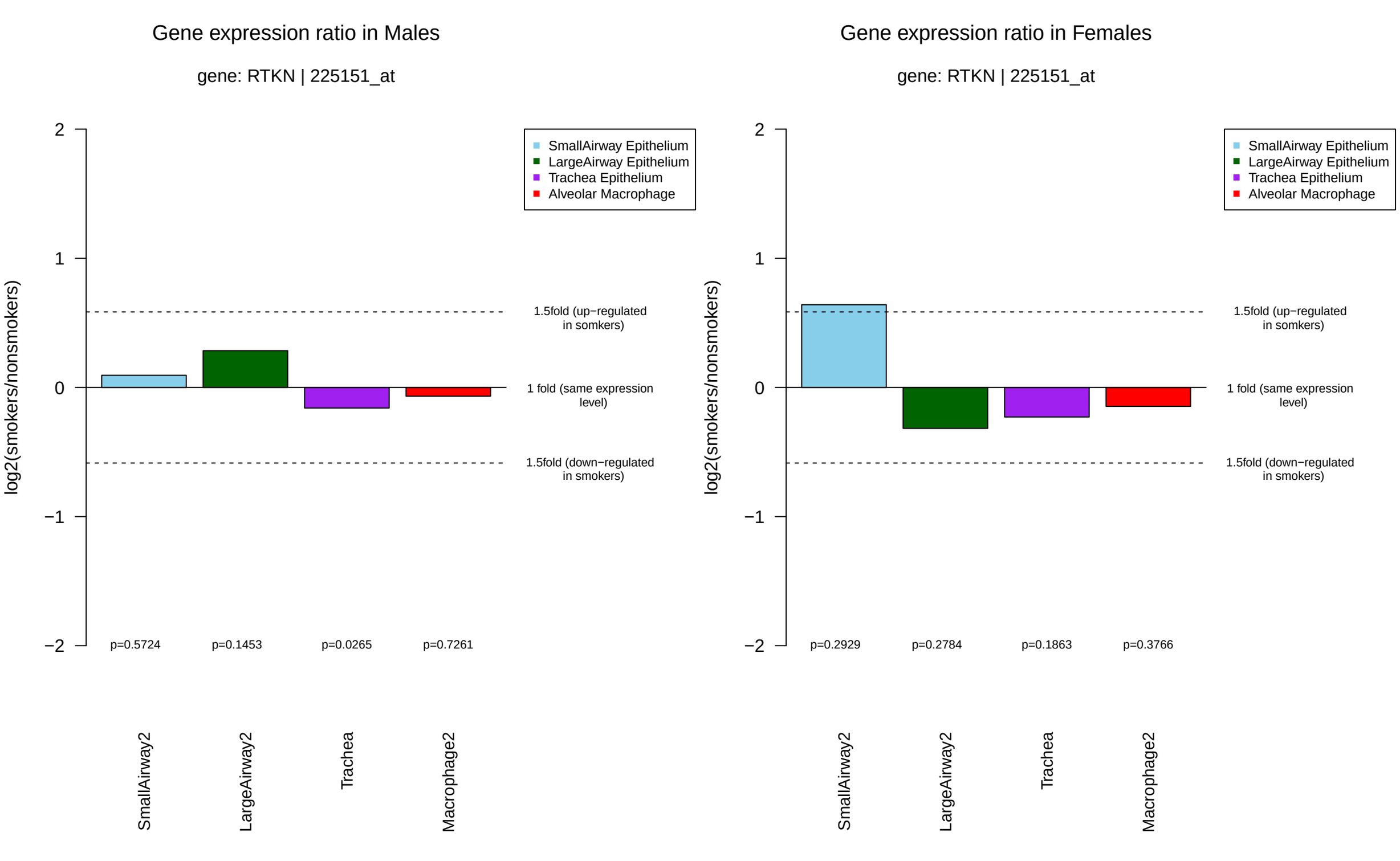
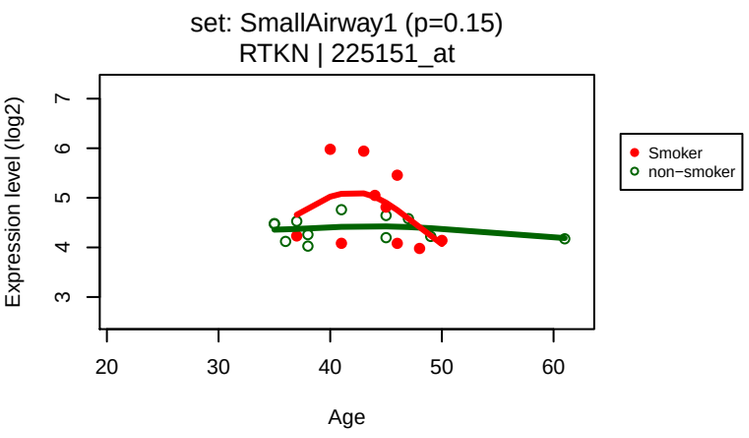
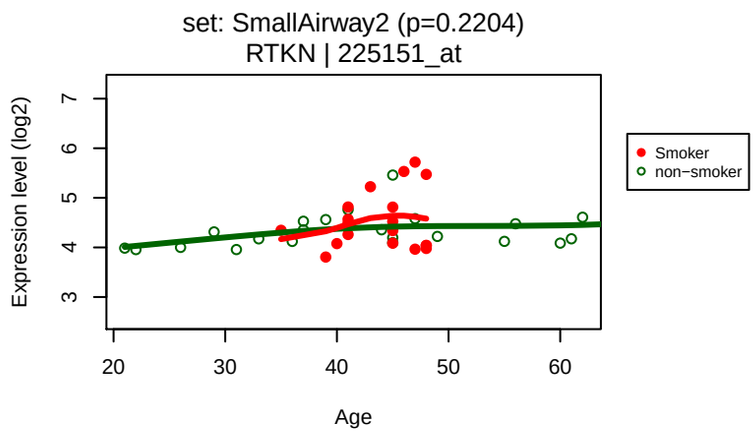
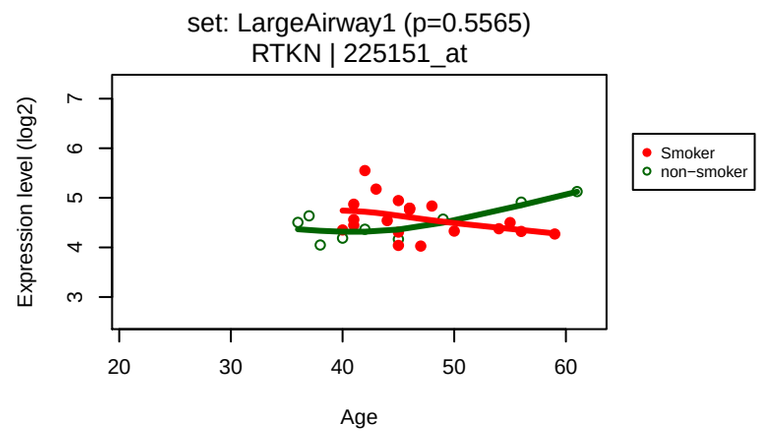

Gene expression of smokers and nonsmokers
gene: RTKN | 225151_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.15
p=0.2204
p=0.3178
p=0.5565
p=0.5911
p=0.0222
p=0.9698
p=0.4715
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

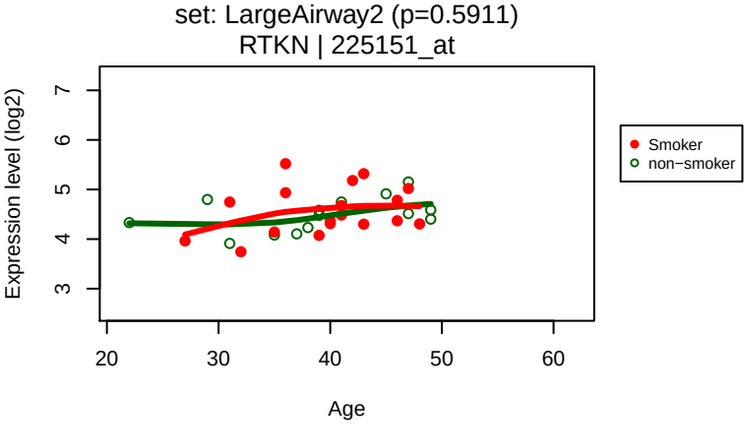

Gene expression ratio (smokers/nonsmokers)
gene: RTKN | 225151_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.15
p=0.2204
p=0.3178
p=0.5565
p=0.5911
p=0.0222
p=0.9698
p=0.4715
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

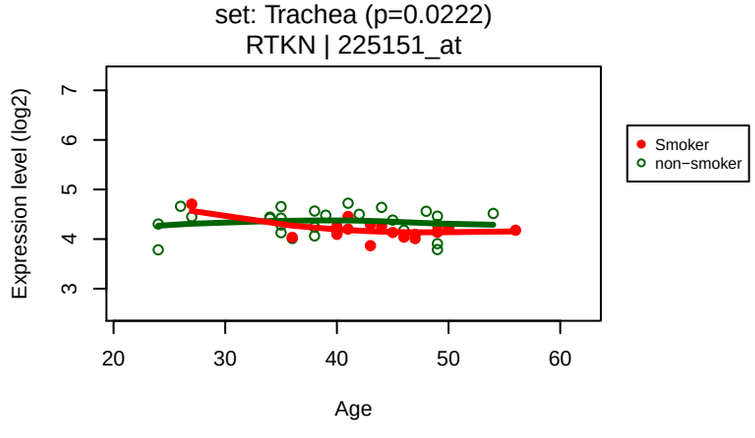

Gene expression ratio in Males
gene: RTKN | 225151_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5724
p=0.1453
p=0.0265
p=0.7261
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RTKN | 225151_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2929
p=0.2784
p=0.1863
p=0.3766
SmallAirway2
LargeAirway2
Trachea
Macrophage2

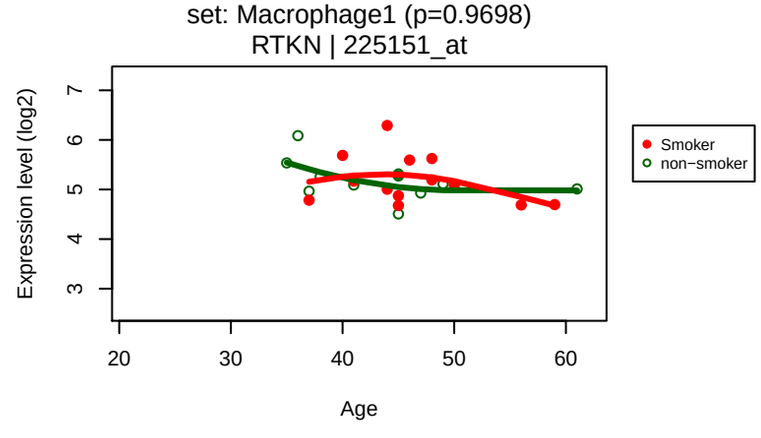

set: SmallAirway1 (p=0.15)
RTKN | 225151_at
Expression level (log2)
Age
Smoker
non−smoker

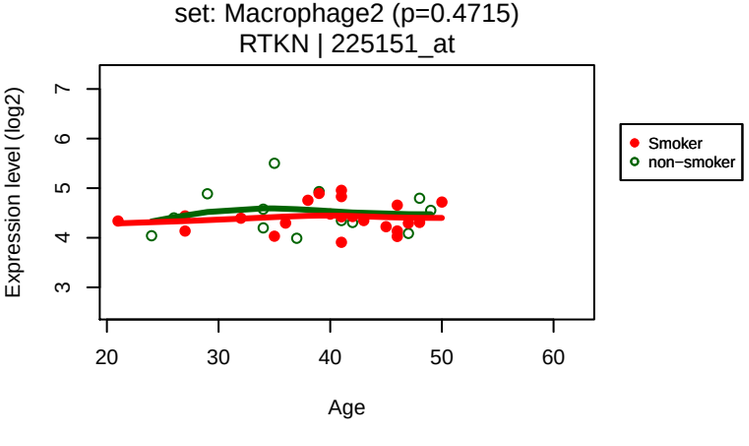

set: SmallAirway2 (p=0.2204)
RTKN | 225151_at
Expression level (log2)
Age
Smoker
non−smoker

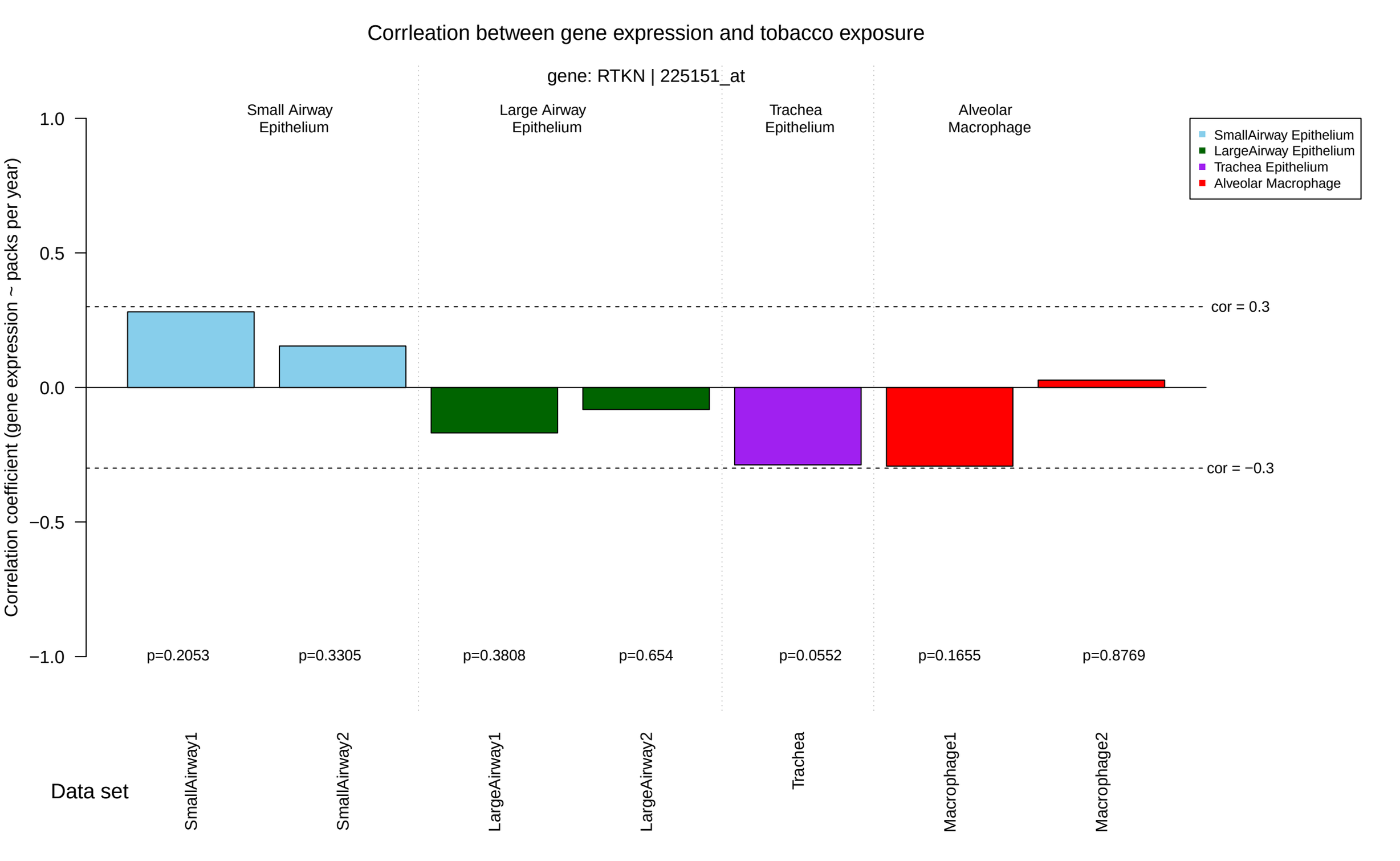

set: LargeAirway1 (p=0.5565)
RTKN | 225151_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5911)
RTKN | 225151_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0222)
RTKN | 225151_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9698)
RTKN | 225151_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4715)
RTKN | 225151_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RTKN | 225151_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2053
p=0.3305
p=0.3808
p=0.654
p=0.0552
p=0.1655
p=0.8769
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2